

Bike Station Overview – Power BI Dashboard

Problem Statement:

Public bike-sharing systems generate continuous data from hundreds of stations across different cities. The task is to analyze this real-time bike station dataset to understand station performance, usage efficiency, and operational patterns across cities.

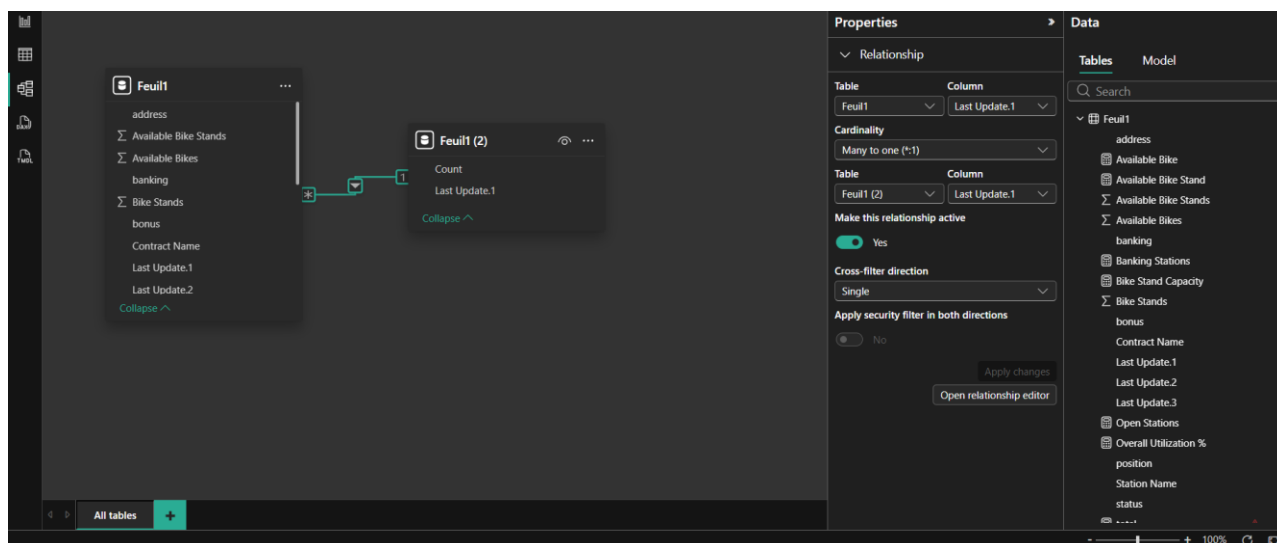
Applying Power BI skills—from data cleaning and modeling to DAX and dashboarding—to turn raw data into actionable urban mobility insights.

Import and export the Power BI

bonus	status	Contract Name	Bike Stands	Available Bike Stands	Available Bikes	Last Update
FALSE	OPEN	seville	20	17	3	
TRUE	OPEN	nancy	25	7	18	
FALSE	OPEN	dublin	31	8	23	
FALSE	CLOSED	vimus	21	0	0	
TRUE	OPEN	nantes	33	28	4	
TRUE	OPEN	creteil	26	26	0	
FALSE	OPEN	valence	25	21	4	
FALSE	OPEN	ljubljana	20	7	13	
TRUE	CLOSED	rouen	25	2	0	
FALSE	OPEN	namur	15	8	7	
FALSE	TRUE OPEN	Bruxelles-Capitale	20	8	12	
FALSE	OPEN	toyama	16	7	9	
TRUE	CLOSED	santander	40	36	4	
FALSE	OPEN	amiens	20	16	0	
FALSE	OPEN	maribor	22	13	9	
FALSE	OPEN	lyon	6	0	0	
FALSE	CLOSED	lillestrom	20	10	10	
TRUE	TRUE OPEN	besancon	30	16	14	
TRUE	CLOSED	mulhouse	40	39	0	
FALSE	OPEN	toulouse	19	1	18	
FALSE	OPEN	luxembourg	20	19	1	
FALSE	OPEN	besancon	10	3	7	
FALSE	OPEN	nancy	16	9	6	
FALSE	OPEN	Bruxelles-Capitale	25	16	7	
TRUE	OPEN	toulouse	24	1	21	
FALSE	OPEN	bruxelles	16	1	9	

address	banking	bonus	status	Contract Name	Bike Stands	Available Bike Stands	Available Bikes	Last Update
GLORIETA OLIMPICA - Aprox. Concejal Alberto Jiménez-Becerril	False	False	OPEN	seville	20	17	3	12 November 2025
CAMINO DE LOS DESCUBRIMIENTOS - Aprox. Calle Isaac Newton	False	False	OPEN	seville	20	0	20	12 November 2025
CALLE AMÉRICO VESPUCIO - Aprox. Calle de Jacques Cousteau	False	False	OPEN	seville	20	15	4	12 November 2025
CALLE AMÉRICO VESPUCIO - Aprox. C/ Gregor Mendel	False	False	OPEN	seville	20	5	14	12 November 2025
CAMINO DE LOS DESCUBRIMIENTOS - Aprox. C/ Albert Einstein	False	False	OPEN	seville	20	1	19	12 November 2025
CALLE JOSE DIAZ - Aprox. Avda Concejal Alberto Jiménez-Becerril	False	False	OPEN	seville	20	19	0	12 November 2025
Avda Alberto Jiménez Becerril - Antes Calle Maimonides	False	False	OPEN	seville	20	6	14	12 November 2025
CALLE DE MANUEL VILLALOBOS - Aprox. C/ Sor Francisca Dorotea	False	False	OPEN	seville	20	13	7	12 November 2025
CALLE DE MADRESELVA - Aprox. Ronda de Pio XII	False	False	OPEN	seville	20	17	1	12 November 2025
AVENIDA EDUARDO DATO - Aprox. C/ Marqués del Nervión	False	False	OPEN	seville	20	15	5	12 November 2025
PARLAMENTO - Aprox. C/ Don Fadrique	False	False	OPEN	seville	20	18	2	12 November 2025
CALLE MANUEL VILLALOBOS - Aprox. C/ Madre San Marcelo	False	False	OPEN	seville	20	16	4	12 November 2025
AVENIDA DE LLANES - Aprox. Avda de Miraflores	False	False	OPEN	seville	20	18	1	12 November 2025
AVENIDA EDUARDO DATO - Detrás Estadio Sánchez Pizjuán	False	False	OPEN	seville	20	20	0	12 November 2025
ALAMEDA DE HERCULES - Aprox. Calle Peris Mencheta	False	False	OPEN	seville	20	10	10	12 November 2025
AVENIDA TORNEO - Aprox. C/ Juan de Acre	False	False	OPEN	seville	20	20	0	12 November 2025
ALAMEDA DE HERCULES - Aprox. C/ Vulcano	False	False	OPEN	seville	20	11	9	12 November 2025
PLAZA SAN ANTONIO DE PADUA - Aprox. C/ Marqués de la Mina	False	False	OPEN	seville	20	15	3	12 November 2025
ALAMEDA DE HERCULES - Aprox. C/ Amor de Dios	False	False	OPEN	seville	20	5	14	12 November 2025
AVENIDA TORNEO - Aprox. C/ Luis Rey Romero	False	False	OPEN	seville	20	18	2	12 November 2025
PLAZA JERONIMO DE CORDOBA- Aprox. C/ Escuelas Pías	False	False	OPEN	seville	20	11	9	12 November 2025
PLAZA DE LA ENCARNACIÓN - Aprox. C/ Santillana	False	False	OPEN	seville	20	8	12	12 November 2025
PLAZA DEL MUSEO - Aprox. C/ Alfonso XII	False	False	OPEN	seville	20	11	8	12 November 2025
CALLE SAN PABLO - Aprox. C/ Bailén	False	False	OPEN	seville	20	17	1	12 November 2025
CALLE ARROYO - Aprox. Avda. José Laguillo	False	False	OPEN	seville	20	9	7	12 November 2025
CALLE LEONARDO DA VINCI - Aprox. C/ Albert Einstein	False	False	OPEN	seville	20	3	17	12 November 2025

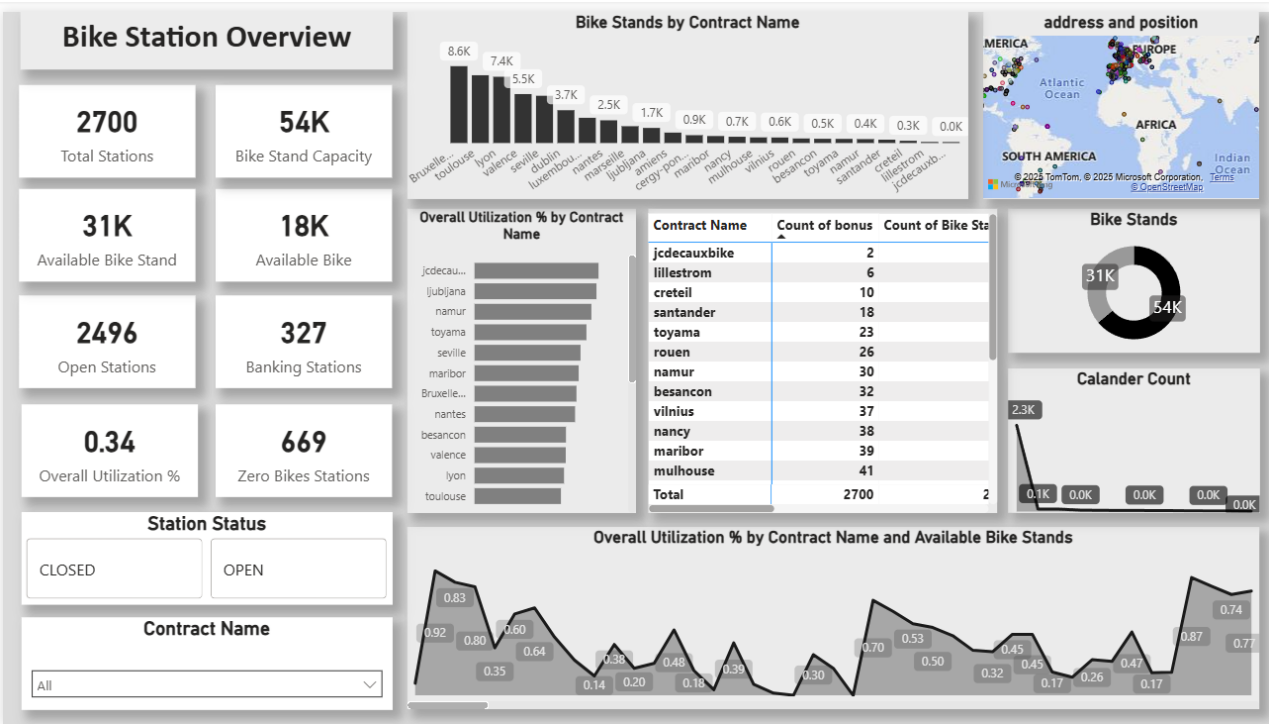
Defining the relationships between the fact and dimension tables



Created a Calculated Columns and DAX Measures

1. Total Stations = COUNT(Feuil1[Station Name])
2. Available Bike = SUM(Feuil1[Available Bikes])
3. Available Bike Stand = SUM(Feuil1[Available Bike Stands])
4. Banking Station = CALCULATE(COUNTROWS('Feuil1'), 'Feuil1'[banking] = TRUE())
5. Bike Stand Capacity = SUM(Feuil1[Bike Stands])
6. Open Stations = COUNTROWS(FILTER('Feuil1', 'Feuil1'[status] = "OPEN"))
7. Overall Utilization % = DIVIDE(SUM('Feuil1'[Available Bikes]),SUM('Feuil1'[Bike Stands]), 0)
8. Total Stations = COUNT(Feuil1[Station Name])
9. Zero Bikes Stations = CALCULATE(COUNTROWS('Feuil1'), Feuil1[Available Bikes] = 0)

Executive Summary Report



Key performance

- **Total Stations – 2700:** Size of the bike station network being monitored.image.jpg
- **Bike Stand Capacity – 54K:** Total number of docking points available across all stations, representing maximum fleet capacity the system can host.image.jpg
- **Available Bike Stand – 31K:** Number of free docks currently available for returning bikes, indicating spare capacity in the network.image.jpg
- **Available Bike – 18K:** Number of bikes ready to use, showing real-time supply available to riders.image.jpg
- **Open Stations – 2496:** Count of stations with status “OPEN”, reflecting the portion of the network that is operational.image.jpg
- **Banking Stations – 327:** Stations that support banking/payment features, useful for understanding payment-enabled coverage.image.jpg
- **Overall Utilization % – 0.34:** System-wide ratio of available bikes to total stands (34%), showing how much of the installed capacity is currently being used.image.jpg
- **Zero Bikes Stations – 669:** Number of open stations with no bikes available, highlighting critical service gaps where users cannot start trips.

Insight and Finding

1. The network is large, with 2700 stations and 54K bike stand capacity, but only 18K bikes are currently deployed, showing spare capacity.
2. Overall utilization is 34%, meaning about two-thirds of docking capacity is not occupied by bikes at this snapshot.
3. There are 2496 open stations, yet 669 have zero bikes, revealing significant service reliability issues and rebalancing needs.
4. Utilization and capacity vary strongly by contract (city), so some cities are using their networks efficiently while others appear under-used.
5. 327 banking stations provide a chance to study whether payment-enabled stations achieve better usage and where to prioritize further upgrades

Conclusion

The dashboard shows that the bike-sharing network has strong infrastructure but is not being used to its full potential, with overall utilization at only 34% despite high station and stand capacity. A large number of zero-bike yet open stations highlights service reliability and rebalancing issues that must be addressed to improve user experience and trust in the system. Differences in utilization across contracts suggest that operational best practices from well-performing cities should be replicated elsewhere, while underperforming areas may need targeted interventions such as redistribution, pricing changes, or capacity right-sizing